

Udayan Shashank Shukla

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EDUCATION

The University of Texas at Dallas, <i>M.S., Business Analytics and Artificial Intelligence</i>	May 2027
National Institute of Technology Allahabad, <i>B.Tech., Mechanical Engineering</i>	May 2023

TECHNICAL SKILLS

Programming & Scripting:	SQL, Python(Pandas, NumPy, Scikit-learn, Matplotlib), Bash,REST APIs
Visualization & BI:	Tableau, Power BI, QlikView, MS Excel (Advanced), Alteryx
Data Platforms:	Snowflake, Databricks, Azure, Microsoft Fabric, AWS (S3, Redshift, Lambda, SageMaker, QuickSight, Redshift, S3, EC2, RDS, DynamoDB),JSON
Big Data & ETL:	Apache Spark (PySpark), Hadoop (HDFS, Hive), dbt, SQL Server, MongoDB
Containers & DevOps:	Docker, Podman, Vector Databases(PineCone, ChromaDB), Graph Databases(Neo4j)
ML & AI:	Machine Learning (Classification, Regression, Clustering), NLP, Sentiment Analysis, AI Model Governance, RAG(retrieval augmented generation),CNN(Convoluted Neural Network)

Work EXPERIENCE

Ernst and Young, Bengaluru, India	
Gen AI Engineer/ Advanced Analyst	Aug 2023 – May 2025
- Collaborated directly with EY partner on a pharma due diligence project; identified missing/skewed datasets using Python, Alteryx, and Excel, improving accuracy of analysis by 25% and reducing validation time by 15 hours. - Automated data validation workflows that reduced manual review time by 40%. - Designed and deployed GenAI-powered solutions leveraging OpenAI APIs, RAG, Vector Databases improving efficiency in client reporting by 30%, and received two Spot Awards for excellence and innovation in analytics projects.	

PROJECTS

Intelligent Multimodal Agent 3D Task Execution: GitHub	
- Built an autonomous agent that interprets natural language commands and performs object manipulation tasks in a simulated 3D environment. - Integrated LLM-based language parsing, deep learning–powered vision, and multi-step action planning using AI2-THOR. - Containerized the system for reproducible experiments and optimized performance with modular, asynchronous architecture.	
Semiconductor Wafer Defect Detection: GitHub	
Engineered a computer vision pipeline utilizing OpenCV and NumPy to parse and normalize over 25,000 raw numeric arrays into high-contrast image datasets for YOLOv8 model training. Trained and deployed a custom YOLOv8 model to identify 8 distinct semiconductor manufacturing defects, achieving a 96.2% mAP50 and 99.4% accuracy specifically on Edge-Ring anomalies. Developed a predictive material waste model using Scikit-Learn’s Random Forest Regressor on a historical database of 823K+ wafers, predicting scrap rates with an MAE of 0.09%. Managed version control via Git, collaborating with team members to integrate a Python-based robotic control middleware and centralized SQLite database for automated defect routing.	

ACHIEVEMENTS AND LEADERSHIP EXPERIENCE

Runner Up – AWS Agentic AI Hackathon 2025, UT Dallas, Texas (Sponsored by AWS)	Oct 2025
2 Spot Awards for due diligence project in healthcare sector at Ernst and Young	Dec 2024